1.A) TE10 Mode Simulation: A WR-90 rectangular waveguide has a = 22.86 mm, b = 10.16 mm and length L = 100 mm. Operate it at f = 10 GHz and excite the dominant TE10 mode in HFSS. Obtain the E-field and H-field distributions, simulated cutoff frequency fc and propagation constant β. Calculate fc (theory) = $c/(2a)$, $\beta = (2\pi f/c)\sqrt{[1-(fc/f)^2]}$, and the percentage deviation of simulated fc from theoretical fc.

CODE:

```
clc;
clear;
close all;

% WR-90 Rectangular Waveguide
a = 22.86e-3;       % Width in meters
b = 10.16e-3;       % Height in meters
L = 100e-3;         % Length in meters

% Operating Parameters
f = 10e9;           % Operating frequency (Hz)
c = 3e8;            % Speed of light (m/s)
mu0 = 4*pi*1e-7;
eps0 = 8.854e-12;

% Theoretical Cutoff Frequency
fc = c/(2*a);

% Free Space Wavenumber
k0 = 2*pi*f/c;

% Propagation Constant
```

```
beta = k0*sqrt(1-(fc/f)^2);

% Display Results
fprintf('\n===== TE10 MODE SIMULATION =====\n');
fprintf('Waveguide Width = %.2f mm\n',a*1000);
fprintf('Waveguide Height = %.2f mm\n',b*1000);
fprintf('Waveguide Length = %.2f mm\n',L*1000);
fprintf('Operating Frequency = %.2f GHz\n',f/1e9);
fprintf('Theoretical Cutoff Frequency = %.4f GHz\n',fc/1e9);
fprintf('Propagation Constant = %.2f rad/m\n',beta);

% Coordinates
x = linspace(0,a,100);
y = linspace(0,b,50);
[X,Y] = meshgrid(x,y);

% TE10 Electric Field Distribution
Ey = sin(pi*X/a);

% TE10 Magnetic Field Distribution
% Normalized field patterns
Hx = -(beta/(2*pi*f*mu0))*Ey;
Hz = -(pi/(2*pi*f*mu0*a))*cos(pi*X/a);

% Plot 1: Electric Field at the Waveguide Center
figure(1);
plot(x*1000,Ey(25,:),'LineWidth',2);
xlabel('Width (mm)');
ylabel('Normalized Electric Field');
title('TE10 Electric Field Distribution');
grid on;
```

```
% Plot 2: Magnetic Field Hx
figure(2);
plot(x*1000,Hx(25,:),'LineWidth',2);
xlabel('Width (mm)');
ylabel('Hx (A/m per normalized amplitude)');
title('TE10 Magnetic Field Hx');
grid on;

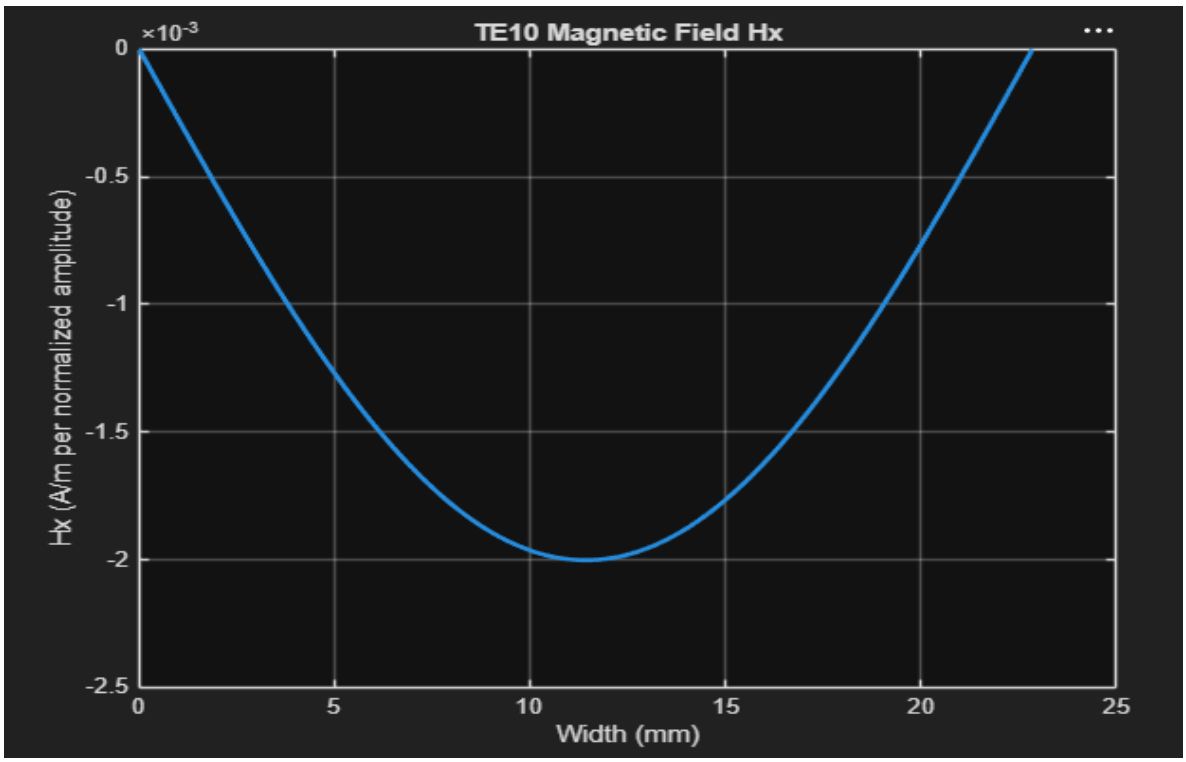
% Plot 3: Magnetic Field Hz
figure(3);
plot(x*1000,Hz(25,:),'LineWidth',2);
xlabel('Width (mm)');
ylabel('Hz (A/m per normalized amplitude)');
title('TE10 Magnetic Field Hz');
grid on;

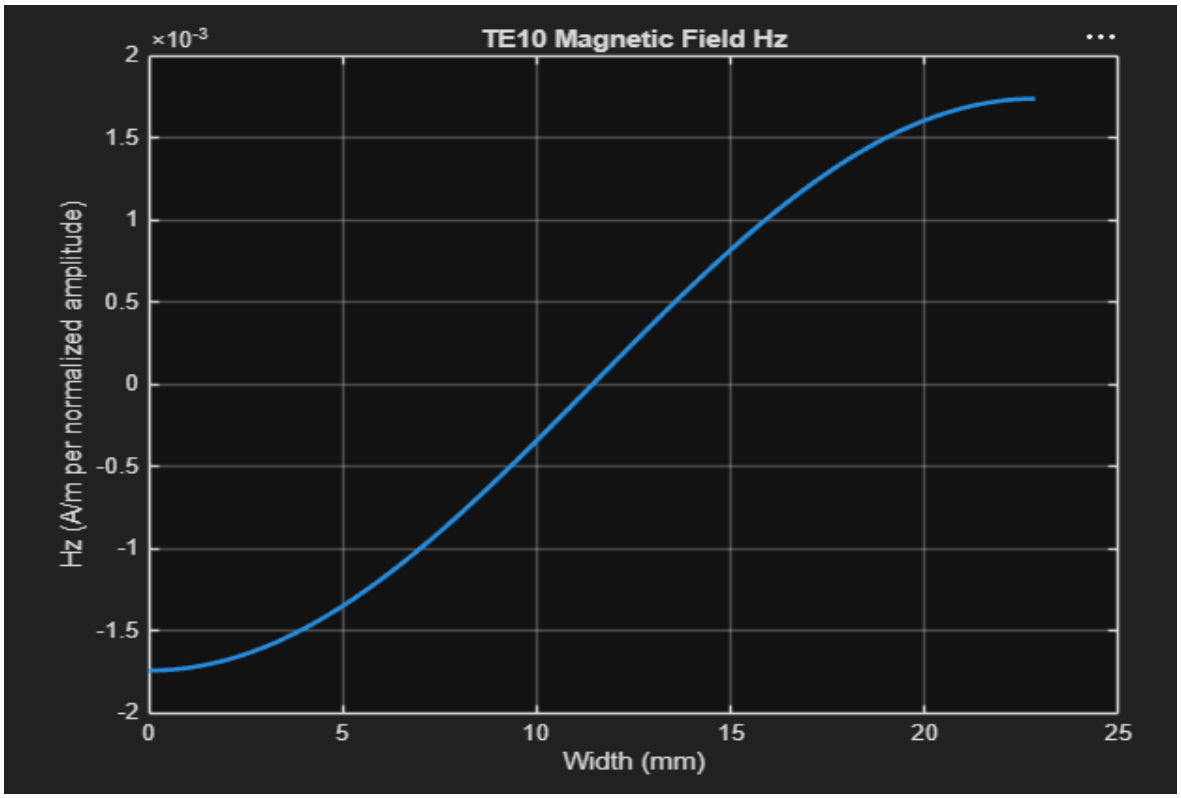
% Percentage Deviation
% Enter the simulated cutoff frequency from HFSS in GHz.
% Replace NaN with your measured value.
fc_sim = NaN;

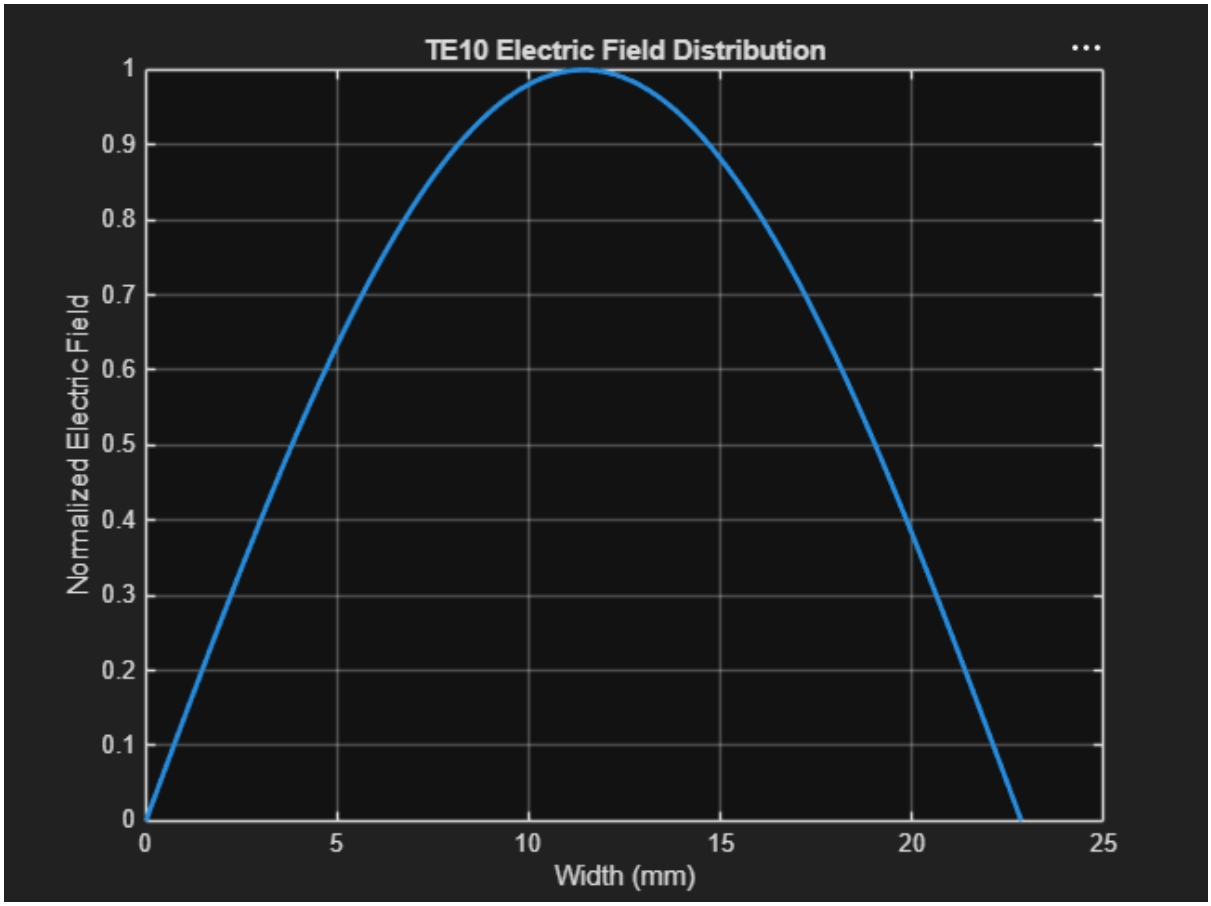
if ~isnan(fc_sim)
   deviation = abs(fc_sim-fc/1e9)/(fc/1e9)*100;
   fprintf('HFSS Cutoff Frequency = %.4f GHz\n',fc_sim);
   fprintf('Percentage Deviation = %.4f %%\n',deviation);
else
   fprintf('HFSS Cutoff Frequency: Not entered\n');
   fprintf('Percentage Deviation: Calculate after HFSS simulation\n');
end

fprintf('\nTE10 mode simulation completed.\n');
```

OUTPUT


TE10 Magnetic Field Hx
×10-3
0
-0.5
-1
-1.5
-2
-2.5
Hx (A/m per normalized amplitude)
0
5
10
15
20
25
Width (mm)


TE10 Magnetic Field Hz
×10-3
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
Hz (A/m per normalized amplitude)
0
5
10
15
20
25
Width (mm)


TE10 Electric Field Distribution
Normalized Electric Field
Width (mm)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
5
10
15
20
25